

The Prosody of Theticity and Focus in Beginner L2 Spanish

Sebastian Leal-Arenas, Marta Ortega-Llebaria

University of Pittsburgh

sal209@pitt.edu, mao61@pitt.edu

Abstract

To express focus on the subject, English speakers make the word in focus perceptually more prominent by enhancing its duration, F0, and intensity, e.g., “ANNE is sleeping”. In contrast, Spanish speakers use verb-subject inversion to express an equivalent meaning, e.g., “Duerme Ana”. Similarly, thematic-categorical meanings are expressed via intonation in English and word order in Spanish. Even at advanced proficiency levels, English learners of Spanish continue transferring their L1 intonation strategy while struggling to use word order to convey those meanings. The present study investigates whether explicit instruction to elementary Spanish learners on the use of word inversion to express subject-verb focus and thematic-categorical contrasts leads to the suppression of English focal prominences, and consequently, to the adoption of Spanish intonation. Thirty learners, divided into experimental ($N = 20$) and control ($N = 10$) groups, completed a pretest, a 10-session training, and a posttest. Participants were asked to produce complete sentences with subjects and verbs. Results revealed that accurate word order improved in the experimental group. Acoustic intonation analysis showed that only those participants with accurate word order and no pauses stopped transferring L1 intonation, approaching Spanish monolingual values. This suggests that, for those learners, word order precedes the learning of Spanish sentence melody.

Index Terms: focus, implicit teaching, L2 intonation, speech production, theticity.

1. Introduction

English and Spanish express equivalent meanings, for instance, subject-focus (1) and verb-focus (2), using different linguistic strategies. Spanish uses word order to highlight the word in focus (underlined word in examples below) by placing it at the end of the sentence. As a result, a verb-subject (VS) word order denotes focus on the subject (1b), and a subject-verb (SV) word-order expresses focus on the verb (2b) (see for instance [4], [5] for dialectal variation to this general pattern). In contrast, English uses prosody to make the word in-focus sound more prominent while maintaining a subject-verb order [1], [2], [3]. Consequently, the word in-focus exhibits longer duration, louder intensity, and larger pitch range than words in post-focal positions. This phenomenon is referred to as *in-situ* prosody.

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|-----|----------------------------|----------------------------|
| (1) | a. ¿Quién llora? | Who’s crying? |
| | b. Llorar el <u>bebé</u> . | The <u>baby</u> ’s crying. |
| (2) | a. ¿Qué hace el bebé? | What’s the baby doing? |
| | b. El bebé <u>llora</u> . | The baby’s <u>crying</u> . |

Similarly, Spanish uses word order and English prosody to express thematic and categorical meanings. Thematic statements refer

to relatively short sentences consisting of a non-pronominal subject followed by an intransitive verb that simply introduces an event or situation [6], [7], [8], [9], [10]. On the other hand, *categorical statements* attribute a property to the subject, where the subject is the ‘topic’ and the predicate the ‘comment’ [11]. In Spanish, these two meanings are expressed with different word orders, i.e., thematic statements present a verb-subject order as in (1) and categorical statements a subject-verb order as in (2). English employs a prosodic strategy where the thematic statements enhance the word in the subject (1), and categorical statements the verb (2).

Shifting from the English prosodic strategy to the Spanish word-order strategy is challenging for English speakers learning Spanish [12], [13], [14], especially for those in the early stages of language acquisition. Results in [12] showed that beginner learners produced little to no VS word order in both sentence types, and only participants with advanced language proficiency started producing VS word order. In [14], expert users of Spanish completed a written preference task. Results indicated that participants used word order as native Spanish speakers to produce the thematic-categorical contrast. However, both SV and VS word orders were acceptable in subject focus statements. Overall, these studies collectively demonstrate the challenge faced by both beginner and advanced English speakers of Spanish in using Spanish’s characteristic word order. Moreover, most research [12], [13], [14], [15] has focused solely on syntactic explanations of thematicity and focus encoding, without considering the impact of prosody.

Teaching intonation poses a challenge. Variations in pitch, stress, and rhythm, which contribute to conveying intended meanings, are frequently subtle and context-dependent. Consequently, learners often find it difficult to consciously manipulate intonation patterns, which are typically acquired through exposure and practice in real communication settings.

Given the above challenges, the present study investigates whether explicit instruction to elementary Spanish learners on the use of word inversion to express subject-verb focus and thematic-categorical contrasts leads to the reduction of *in-situ* prominences, and consequently, to the adoption of Spanish intonation. The following research questions guided this study:

RQ1: What is the effect of explicit training on word order in the production of thematic and subject focus statements by non-expert users?

RQ2: Does explicit word order training change *in-situ* English prosody in Spanish?

ratios as dependent variables, and test, group, and meaning with their interactions as fixed factors, and participants and sentences as random effects were performed. Fixed effects variables were contrast coded.

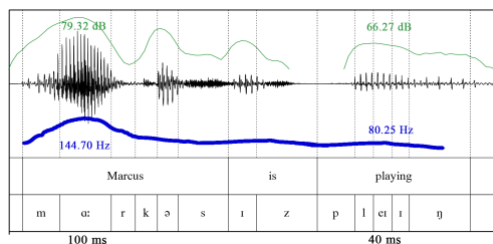


Figure 3a: English sentence 'Marcus is playing'. Pitch accent on 'Marcus'. Intensity (dB) is represented as a green line. F0 trajectory (Hz) is the blue line. Duration (ms) values provided under the stressed syllable in each word.

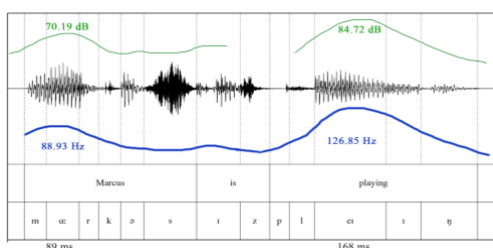


Figure 3b: English sentence 'Marcus is playing'. Pitch accent on 'playing'. Intensity (dB) is represented as a green line. F0 trajectory (Hz) is the blue line. Duration (ms) values provided under the stressed syllable in each word.

3. Results

3.1. RQ1: Subject-verb inversion

The experimental group presented an increase in syntactic accuracy in thetic (20%; carried by 4 participants) and subject focus (40%; carried by 8 participants) statements. However, syntactic accuracy was the same for the control group in both test types, with no instances of SV inversion as shown in Table 1. There was no inversion in categorical or verb focus statements.

Table 1: Subject-verb inversion results. Thetic and subject focus sentences present syntactic accuracy by SV inversion (I). Categorical and verb focus sentences do not (NO).

Sentence	Experimental		Control	
	Pretest	Posttest	Pretest	Posttest
Categorical (NO)	100%	100%	100%	100%
Thetic (I)	0%	20%	0%	0%
Verb Focus (NO)	100%	100%	100%	100%
Subject Focus (I)	0%	40%	0%	0%

The mixed-effect model showed significant results for meaning ($\beta = -21.6$, $p < .001$), test type * group ($\beta = 3.35$, $p < .001$), and group * meaning ($\beta = -1.69$, $p < .001$) confirming that training caused the control group to start using the syntactic strategy.

3.2. RQ2: Phonetic correlates to phrase-level accentuation

The second research question examines whether participants in the experimental group ceased using the English *in-situ* prosodic strategy when speaking Spanish. Figures 4-6 indicate that some individuals in the experimental group did stop using it after training. In the pretest, duration ratios (Figure 4) showed comparable scores for the experimental and control groups, both above 1 for the thetic statements (blue dots) and subject-focus (grey), and below 1 for the categorical (yellow) and verb-focus (green) statements, indicating the use of English *in-situ* prosody. However, in the posttest, only the experimental group showed reduced ratio scores in thetic (grey) and subject focus (blue) utterances, reaching values below 1. This demonstrates a reduction in the use of English *in-situ* prosody after training. Similar patterns are observed in Figure 5 for intensity. Figure 6 illustrates a reduction in F0 values in the experimental group without reaching values below 1.

Results from the mixed models revealed statistically significant results for duration (meaning: $\beta = 2.70$, $p < .001$, test * meaning: $\beta = 3.671$, $p < .001$, group * meaning: $\beta = -2.586$, $p < .001$), marginally significant for intensity (test : $\beta = 50.55$, $p .062$, and did not reach significance for F0).

4. Discussion

The experimental group's improvement in SV inversion in thetic (20%) and subject focus (40%) statements is a notable outcome. Previous research [12], [13], [14], [21], [22] has characterized low-proficiency speakers as unable to use VS order to express theticity and focus. However, we have shown that these learners can use syntax to express meaning. That VS word order was present in more subject focus than thetic statements could be due to the salience of focus expression [23], [24], [25]. Participants may have used their L1 salience cues as a resource for identifying L2 subject-focus sentences, which for Spanish would not be effective. Additionally, participants did not overgeneralize the word order of thetic and subject focus sentences to other sentence types, providing evidence of their understanding of lexical-semantic features of the language, despite their limited language exposure.

Regarding the implicit effect of teaching word order on duration, intensity and F0 peak, results are encouraging. While all three correlates to phrase-level accentuation presented comparable patterns in mean scores across groups, test and sentence types, changes in the experimental group can be accounted for. The variation exhibited in the posttest is carried by those participants who accurately used VS word order in thetic and subject focus statements. Additionally, the number of Intonation Phrases (IPs) in each utterance also influenced duration and intensity. On average, participants who uttered thetic and subject focus statements with 2 IPs continued having values above 1, although these values are lower compared with mean values in the pretest. In statements with 1 IP, mean values were below 1 for duration in thetic ($M = 0.93$; $SD = 0.313$) and subject focus ($M = 0.86$; $SD = 1.588$) statements. The same occurred in intensity values. The tense participants used affected their performance. Those who transferred prosodic features into their L2 used the present continuous. In contrast, those whose duration, intensity and F0 peak are closer to native Spanish, used the present simple and rarely the present continuous in the posttest.

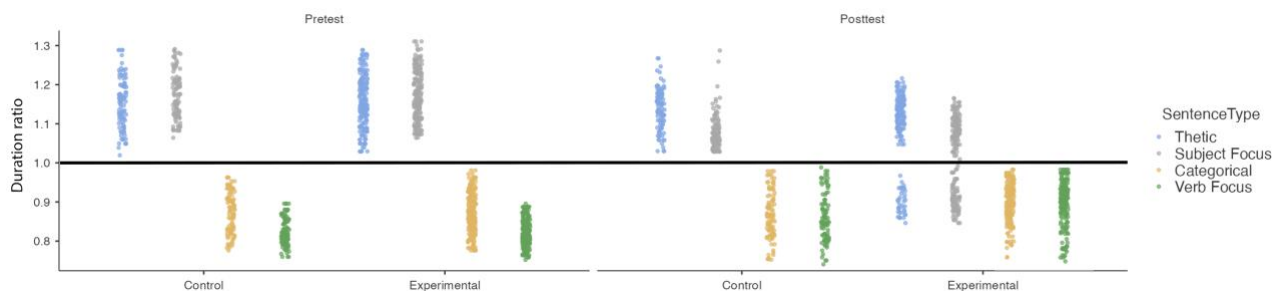


Figure 4: Duration ratios for sentence type, group and test instance. Values above 1 indicate English phrasal accentuation.

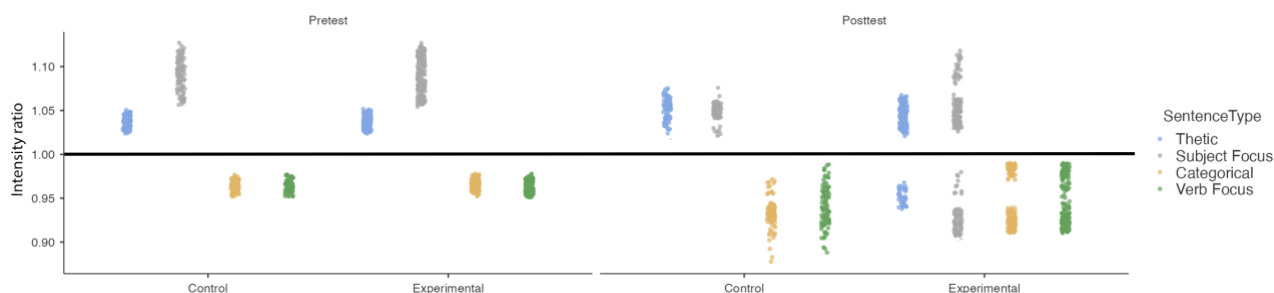


Figure 5: Intensity ratios for sentence type, group and test instance. Values above 1 indicate English phrasal accentuation.

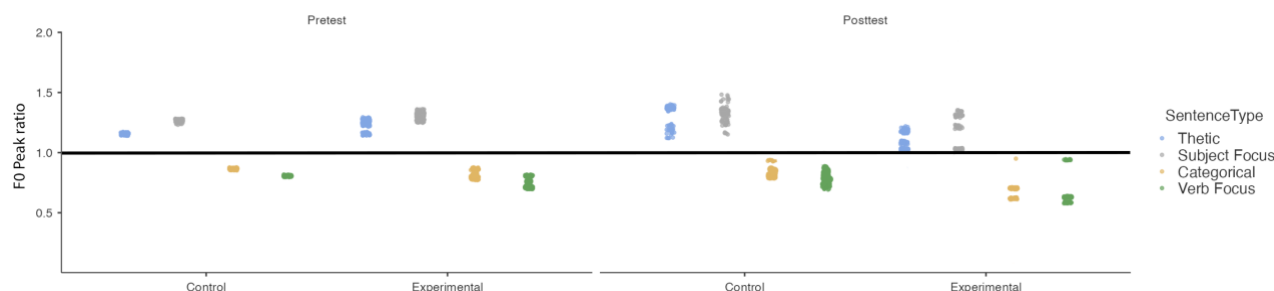


Figure 6: F0 peak ratios for sentence type, group and test instance. Values above 1 indicate English phrasal accentuation.

Participants who got closer to the expected values in Spanish sentences exhibited a higher degree of fluency compared to the other participants in the posttest. Their average sentence length was 300 to 500 ms shorter compared to that of other participants. [22] observed that individuals with limited L2 proficiency exhibited disfluent speech, including pauses and stuttering. In the present study, most participants uttered sentences with 3 IPs. The creation of 3 IPs by the participants had a lengthening effect on the words before the pitch accent. Duration was subject to domain-final lengthening [26], [27], reducing the duration of previous syllables. The apparent fluency exhibited by some participants may have been the result of previous language experiences. Nevertheless, the information in their linguistic background questionnaire did not differ from the mean. Their performance in the language proficiency test was also average. No factors suggest an advantage over their peers. Therefore, it is plausible to infer that the training intervention influenced their posttest performance. In a future study, we analyse the performance of expert users to shed light on the different stages of prosody acquisition.

5. Conclusions

The present work contributes new empirical evidence to the study of theticity and focus realizations in L2 Spanish by L1 English speakers. It fills a void in prior research by examining the production of theticity and focus by English learners of Spanish, both before and after undergoing a pedagogical intervention. The results indicate that participants are able to learn word order to convey meaning; however, the impact of training on prosody was only noticeable in those participants who accurately used verb-subject order in thetic and subject focus statements. Among the prosodic cues, duration was the only one that showed statistically significant changes in the expected direction. While intensity and fundamental frequency (F0) also exhibited changes in the expected direction, these changes did not reach statistical significance. Therefore, it seems reasonable to suggest that syntactic learning precedes the development of prosodic competence.

6. References

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